



Society of Petroleum Engineers

SPE-230107-MS

Breaking the Flow Barrier: An Innovative Data Driven Chemical Solution to Revive Failing Injection Wells in Oman

C. Okocha, A. R. Thornton, and M. Ossama, Clariant Oil Services, Houston, Texas, United States of America

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/230107-MS](https://doi.org/10.2118/230107-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Water management in remote, arid oilfield locations is critically important for several reasons, including preserving scarce resources, reducing operational costs, ensuring regulatory compliance, and minimizing environmental impact. In areas, where water is a precious commodity, efficient water management is essential for sustainable and profitable oil and gas operations. Re-purposing produced water for re-injection into the reservoir for pressure maintenance and sweep efficiency ticks many of the desirable water management boxes, however this can come at a price, as produced water reinjection (PWRI) is often accompanied by troublesome flow assurance and injectivity challenges due to the complex nature of the salt-rich brine mixture.

The study oilfield in Oman employs PWRI for reservoir pressure maintenance and waterflood sweep efficiency. The highly saline field water is generally calcium ion rich and contains other significant multivalent divalent scaling species and suspended inorganic particulates. The oilfield has an ongoing technical program for maximizing production which has yielded impressive results as each successive new production enhancement technology has been introduced, with further enhancements planned across the next 5 year period. Key to the successful exploitation of the 500+ wells draining the source Shuaiba reservoir are introduction of vertical and then horizontal waterflooding technologies, and an extensive ongoing reservoir flow path re-engineering and restructuring program using multiple relative permeability modification (RPM) and water shut-off (WSO) treatments. The next planned optimization project required water injection to increase from 10k to 20k BWPD and the field PWRI system is expected to play a key role in achieving this. The PWRI array performance to-date has been sub-optimal and this is mainly due to periodic well injectivity loss due to solids fouling issues occurring topside and downhole. An Injectivity Improvement taskforce was created to resolve the PWRI injectivity issues and meet the desired current and future injector uptime/injection performance targets. The project group initiated a 2-phased response to the injectivity loss issue;

Phase (I) – comprised a comprehensive data-driven system review of the produced water system in conjunction with an intensive sampling and analysis program targeted at sensitive characterization of recovered fouling solids in the injection array via SEM, ICP and XRD.

Water management in remote, arid oilfield locations is critically important for several reasons, including preserving scarce resources, reducing operational costs, ensuring regulatory compliance, and minimizing

environmental impact. In areas, where water is a precious commodity, efficient water management is essential for sustainable and profitable oil and gas operations. Re-purposing produced water for re-injection into the reservoir for pressure maintenance and sweep efficiency ticks many of the desirable water management boxes, however this can come at a price, as produced water reinjection (PWRI) is often accompanied by troublesome flow assurance and injectivity challenges due to the complex nature of the salt-rich brine mixture.

The study oilfield in Oman employs PWRI for reservoir pressure maintenance and waterflood sweep efficiency. The highly saline field water is generally calcium ion rich and contains other significant multivalent divalent scaling species and suspended inorganic particulates. The oilfield has an ongoing technical program for maximizing production which has yielded impressive results as each successive new production enhancement technology has been introduced, with further enhancements planned across the next 5 year period. Key to the successful exploitation of the 500+ wells draining the source Shuaiba reservoir are introduction of vertical and then horizontal waterflooding technologies, and an extensive ongoing reservoir flow path re-engineering and restructuring program using multiple relative permeability modification (RPM) and water shut-off (WSO) treatments. The next planned optimization project required water injection to increase from 10k to 20k BWPD and the field PWRI system is expected to play a key role in achieving this. The PWRI array performance to-date has been sub-optimal and this is mainly due to periodic well injectivity loss due to solids fouling issues occurring topside and downhole. An Injectivity Improvement taskforce was created to resolve the PWRI injectivity issues and meet the desired current and future injector uptime/injection performance targets. The project group initiated a 2-phased response to the injectivity loss issue;

Phase (I) – comprised a comprehensive data-driven system review of the produced water system in conjunction with an intensive sampling and analysis program targeted at sensitive characterization of recovered fouling solids in the injection array via SEM, ICP and XRD.

Phase (II) - focused on remediation measure implementation involving both PWRI well intervention acid stimulation treatment, and simultaneously topsides produced water chemical treatment strategy options.

PWRI well deposit sample analyses revealed calcium carbonate (CaCO_3) and calcium sulfate anhydrite (CaSO_4) scales as the primary scales of concern across the PWRI array, with some iron sulphide presence; these were responsible for ‘black water’ formation during produced water sampling. Early PWRI CT mineral acid stimulations did not achieve lasting injectivity improvement, however deposit analyses data from recovered samples allowed reformulation of the stimulation package and resulted in stimulation treatments that showed significant and sustained injectivity improvement even though the well pressure itself rose quickly post-stimulation.

Topside produced water flowlines upstream of the PWRI array showed visible scale formation and these particulates were being directed into the PWRI wells. Besides blocking the near wellbore formation downhole, the scale also manifested as a dark colored blocking deposit topsides at the PWRI pump inlet filters, which required regular cleaning to free up. Uninhibited scale formation occurring in the topsides system was believed to be responsible for worsening even marginal scaling tendencies downstream by providing ideal low energy heterogeneous pathways for further scale formation.

A combined approach to scale mitigation was agreed and initiated involving;

- i. Application of FeS dissolver at the main PWRI header to dissolve/loosen existing FeS from legacy accumulations
- ii. Periodic PWRI Header and Flowline flushing to remove accumulated solids until clean water is confirmed
- iii. Deployment of a scale inhibitor topsides for mitigation of CaCO_3 and CaSO_4 scales
- iv. Routine backflushing of injector well flowlines.

Results after six months execution of the Study PWRI particulate mitigating strategy:

- Significant PWRI injectivity improvement allowing design targets to be met.
- Overall field production uplift (based on the study PWRI well test data).
- Reduction in number of PWRI acid stimulation interventions.
- Reduced OPEX due to implementation of the proactive, preventive chemical integrated treatment program.

The success of the combined approach demonstrated the effectiveness of close cooperation between internal project groups and 3rd party service providers in developing a targeted, data-driven PWRI scale control strategy for resolving the complex subsurface flow assurance challenges and optimizing field production. The integrated systematic chemical and operational approach demonstrated measurable success in restoring injectivity and stabilizing pressure support, showcasing the value of targeted diagnostics and adaptive treatment strategies in complex waterflood environments.

Introduction

The scarcity of water in remote hot oilfield locations demands that all existing potential water resources are used optimally to obtain maximum benefit. Harnessing oilfield produced brine for reinjection into the reservoir for pressure maintenance (PWRI) is one such example, however its use is often accompanied by process and injectivity issues as the highly complex salt-rich produced water mix presents water management challenges. Note: in the following paper all analyses data and photographic images are kindly supplied by study operating company.

The study field is located in Block 5 in the north-west of the Sultanate of Oman, see [Figure 1](#). The complex carbonate Upper Shuaiba Member (USM) provides the bulk of oil production, and is subdivided into A, B, C, D and E sub-layers, with layers D and E being the most prolific. The USM reservoir is renowned for its heterogeneity, which is characterized by multiple ultra-thin carbonate layers of thinly bedded bioclastic packstones and wackestones with bimodal/trimodal pore systems (matrix porosity, vugs, fractures). The multi-thin layer structure complicates injectivity behavior and reduces conformance of some injected fluids. [Kazmi et al. \(2024\)](#). The Shuaiba carbonate reservoirs are generally low permeability, however they have extensive natural fracture network patterns that create high-permeability fluid channels that can lead to injected waters bypassing oil-bearing zones resulting in poor sweep efficiency. In Study, many producers using horizontal line-drive waterflooding schemes consequently show high watercuts (70 – 90%) [Zhang et al. \(2007\)](#).

Early field development featured vertical producers operating in depletion mode. Introduction of waterflooding provided new life and improved oil recovery from the producers typically configured with downhole Sucker rod Pumps (SRPs) or Electro-submersible Pumps (ESPs). The initiative has since been enhanced by horizontal waterflooding for pressure support and sweep efficiency specifically targeted at the field horizontals. Of equal importance however was the ongoing long-term field-wide reservoir fluid flowpath restructuring and re-engineering program (2010 – 2025) using state of the art RPM and WSO technologies to obtain more effective waterflood sweep efficiency in the fractured carbonate and realizing the potentials for all field producers. [Al-Mugheiry et al. \(2007\)](#) and [Al-Mufargi et al. \(2025\)](#). New horizontal well designs featuring relocated SRPs deeper in the well - away from the vertical section, will further improve performance and uptime as trial pilots showed the new configuration to be significantly less prone to high gas interference effects. [Al-Harrasi et al. \(2025\)](#). The study field reservoir has limited in-situ natural aquifer support, therefore waterflooding for secondary recovery is crucial to oil and gas recovery from the reservoirs. [Al-Mandhari et al. \(2025\)](#). The improvements detailed above have steadily optimized the efficiency and functionality of the study waterflood, and RPM and WSO re-sculpting of the reservoir fluid

flow network has demonstrably improved subterranean fluid distribution and oil recovery efficiency, and the next planned enhancement project required field water injection to increase further from 10k to 20k BWPd, and the PWRI array plays a key role in achieving this. Al-Mandhari et al. (2025). The PWRI wells have historically demonstrated inconsistent injectivity performance and therefore required urgent review and correction. The project Integrated Collaboration Framework Team project group comprising E&D, Well Engineering, and Field Production expertise were tasked to identify the causes of inconsistent injectivity and then initiate remedial measures to improve performance and reliability. The Injection Improvement Project team initiated a 2-phased response to the PWRI well injectivity issue:



Figure 1—Location of the Oil Field in NW Oman

Phase (I) - included a comprehensive data-driven field produced water system review in conjunction with a deposit sampling and analysis program targeted at sensitive characterization of recovered fouling solids in the injection array, using SEM, ICP and XRD analytical tools.

Phase (II) - focused on remediation measure implementation involving both PWRI well Coiled Tubing (CT) deployed acid stimulation treatment, and topsides produced water line particle flushing and scale inhibitor deployment options, focusing on PWRI wells, INJ-A1, INJ-A2, INJ-A3 and INJ-A4.

PWRI Well Fouling

There are three fundamental well near-wellbore formation blocking mechanisms, although one or more mechanism may occur simultaneously.

Physically induced injectivity decline - due to migration of formation fines and other suspended inorganic particles in the reservoir matrix resulting in flow channel blockage. Also included are internal particle/fines displacement as well as clay swelling (combination of chemical and physical mechanisms) Moghadsil et al. (2004) Feia et al. (2015) Hua et al. (2010), Wang. et al. (2012).

Biologically induced decline - due to bacterial activity, usually occurring in the injector well near-wellbore area where biofilm development occurs impairing fluid flow. Song et al. (2020), Rinck-Pfeiffer (2000) Bennion (2002) Baveye et al. (1998).

Chemically induced decline - due to chemical reaction occurring within the near wellbore formation principally due to mineral scale precipitation. Wang et al. (2018), Chen et al. (2023).

Study PWRI injector wells decline was believed primarily due to suspended inorganic solids transferring from the surface produced water process and passed downhole into the injector well near wellbore formation, and/or scale particles forming during the same fluid journey from surface produced water process to downhole injection and packing at the tight carbonate formation face.

Investigation of fluid samples recovered from a water injector flush – backflow, appeared dark colored or ‘blackish’, which is a typical presentation when iron sulphide is present, see Figure 2. Application of FeS dissolver chemical rapidly removed the dark tinge leaving a light whitish/light yellow hazy solution with sulphate scale particles evident on the base, see Figure 2.

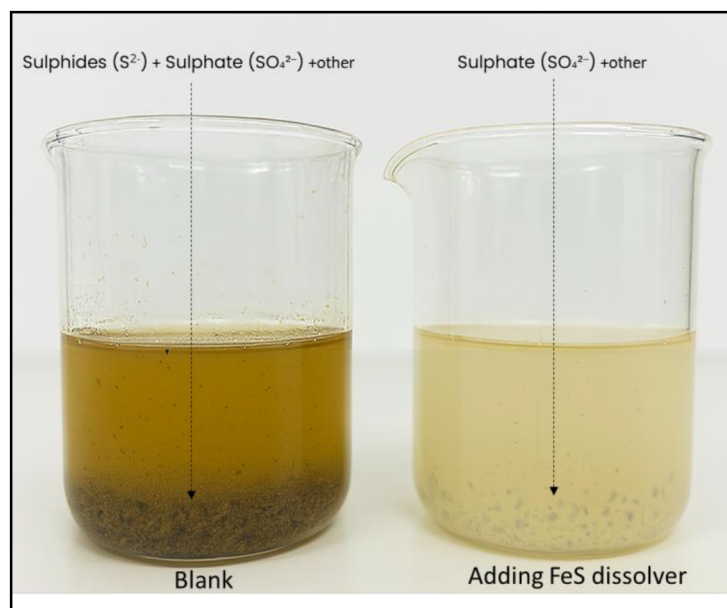


Figure 2—Sample of flowback water returned after flushing a water injector flowline

The acid stimulation treatments deployed into the PWRI to remove blockages only achieved partial and temporary success; the treatments dissolved the acid soluble scales and some carbonate formation rock around the lodged particles allowing higher fluid transmission to occur for a short period of time.

PWRI Well INJ-A1

Decline in PWRI well INJ-A1 injectivity is due to the injected produced water suspended solid load and also produced water scaling occurring as a consequence of mixing of multiple produced waters from a large number of water sources in the surface process. Precipitation of inorganic scale is always a concern when injecting oilfield brines with high concentrations of divalent scaling ions.

A solid sample recovery exercise was performed on well INJ-A1 with samples recovered from topside - at the injection pump inlet filter, and downhole via a ‘Jet Junk Basket’ tool, before and after application of a mutual solvent wash. The sample recovered from downhole via the ‘basket’ is shown in Figure 3.



Figure 3—Sample of debris collected from INJ-A1 downhole via ‘Jet Junk Basket’ tool

The samples were analyzed via XRD for crystal speciation and XRD quantification of crystalline morphologies present in Weight %, see Table 1.

Table 1—XRD of solids recovered from topside and downhole in INJ-A1

Sample Date	Location	Before or After MS slug	Quartz (SiO ₂) Wt%	Anhydrite (CaSO ₄) Wt%	Calcite (CaCO ₃) Wt%	Maghemite (Fe ₂ O ₃) Wt%	Hematite (Fe ₂ O ₃) Wt%	Magnetite (Fe ₃ O ₄) Wt%	Halite (NaCl) Wt%
30 May 24	Surface	Before	45.9	26.2	6.0	9.2	6.3		6.6
30 May 24	D/H Basket	Before	8.3	7.3	70.7			7.0	6.9
31 May 24	Surface	After	13.2	28.6	37.2	8.7	7.5		4.8

**All data supplied by Operator*

The XRD result clearly indicated the deposit sample comprised formation debris (quartz and calcite particles), corrosion by-products and scale particles (CaSO₄ and CaCO₃) which were bound together in a sticky FeS mass, see Figure 3 and Table 1. The presence of halite in the samples may be due to the presence of dehydrated brine salts. The ‘before’ and ‘after’ comparisons for the topside samples really only show that the same crystalline types are present as compared to the downhole sample compliment. It is interesting to note that solubility studies performed on all three deposit samples showed that significant proportions were soluble in both hexane and xylene solvents, suggesting they were hydrocarbon in origin and providing additional information to build up the picture of a developing fouling mass of inorganic particulates coated in crude.

Topside and downhole pressures and injectivity data for three stimulation treatments performed on PWRI INJ-A1 are presented, see Figure 4. The first ‘conventional’ mineral acid CT- deployed stimulation job performed in July-23 generated a reasonable well near-wellbore cleaning response, and provided approximately 3 months injectivity improvement. The February 2024 repeat treatment on INJ-A1 did not achieve the same outcome and ‘provided no clear evidence of PWRI well injectivity improvement’. The 3rd CT stimulation treatment in the series employed the revised optimized recipe (redesigned based on the deposit analyses results) and executed in May 2025. The treatment was considered successful, and achieved 5 months of sustained injectivity (at latest data point), even though downhole pressure was observed to increase quickly; it did not actually impact injectivity performance, see Figure 4.

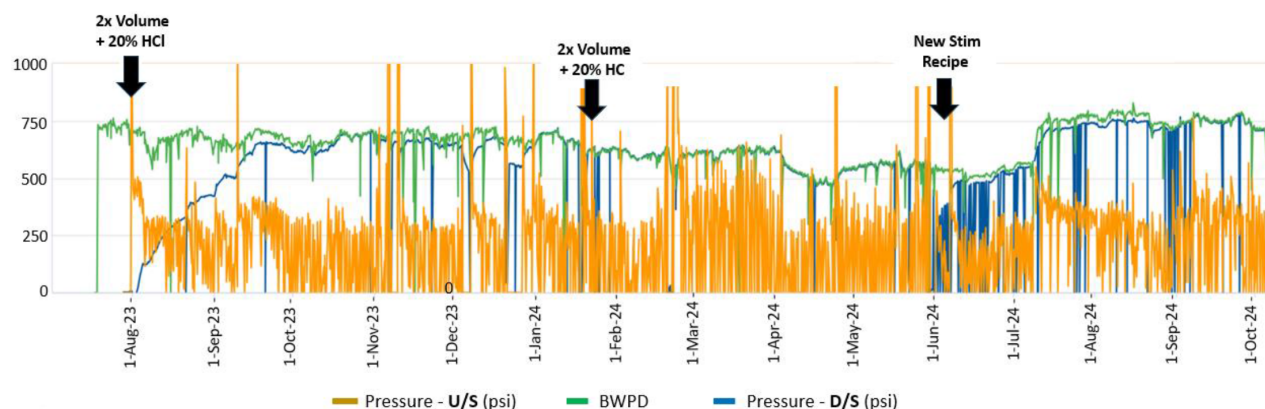


Figure 4—Impact of successive INJ-A1 PWRI acid stimulation treatments from 2023 - 2024

Initiation of Flowline Flushing

Following MOC approval, the project team initiated 10" Main Header flushing from the main water source 'B' station operation in October 2024. All injectors connected to a manifold interface were shut-in, and each PWRI support flowline including INJ-A1, INJ-A2, INJ-A3 and INJ-A4 were flushed for 24 hours using FeS dissolver solution.

All PWRI wells were returned to injection apart from well INJ-A1, which was prepared for October 2024 CT stimulation using the optimized stimulation recipe. Sulphate scale inhibitor dosing commenced into each injector feed water for a trial 3-month period to assess its performance and to determine the optimum dosing location. Ultimately the INJ-A1 injector would act as key guide with respect to assessing the success or failure of the injectivity enhancement actions that were implemented as per the project plan. The INJ-A1 PWRI well injectivity would be closely assessed across the proceeding 6-month period, assessing injection performance bi-weekly to ensure clean water injection and, if necessary, recommend repeat flowline flushing.

A pictorial account of Well INJ-A1 flowline flushing post-iron sulphide dissolver treatment is presented (March 2025), see [Figure 5](#).

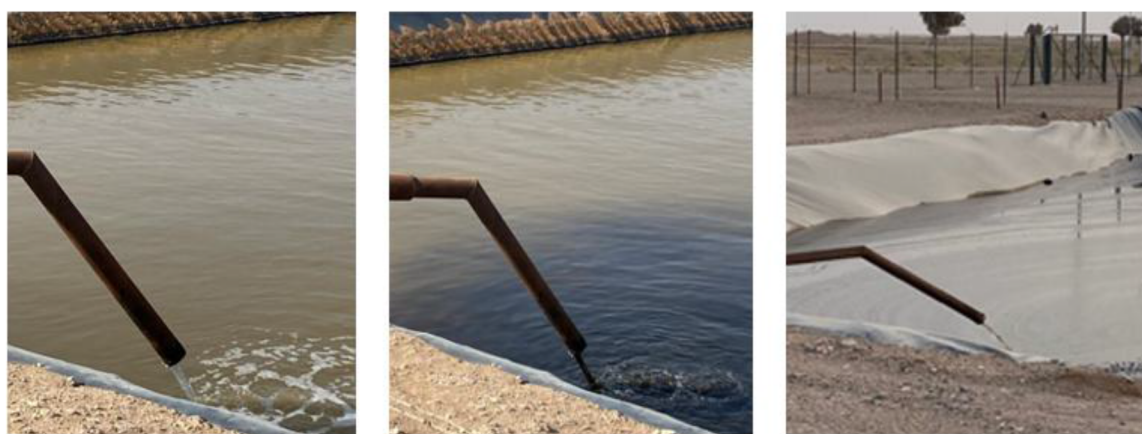


Figure 5—INJ-A1 -021 Flowline Initial Backflush (left), After 10 minutes (middle) and Flush end (right)

Flushing operations commenced at 09:29 on 23 March 25 and returning fluids were initially clear, see left-hand image in [Figure 5](#). After S(see middle image in [Figure 5](#)). The flushing operation was halted at 15:34 when discharging fluids were visibly continuously clear, see RH-image, [Figure 5](#).

PWRI Well INJ-A1 Injectivity performance

The INJ-A1 injector was monitored post-CT-optimized acid stimulation performed on 30th October 2024 (2nd treatment of this type) to assess the impact of subsequent flushing and scale inhibition operations on the well. The INJ-A1 upstream and downstream pressures profiles as well as the well injectivity rate with time are presented, see Figure 6. First flush occurred in November 2024 while scale inhibitor injection commenced into flowline following the CT-stim. All flushing operations are identified, as are the flushing and backflow treatments. It can be seen that flushing operations result in lowering of the D/S pressure initially before it slowly increases indicating increasing resistance at the injection face due to particulate accumulation. Successive flushes appear to have a cumulative effect with injectivity improvement evident. A 3 week period commencing May 2025 had no flushing operations and injection rate was observed to remain stable even though D/S pressure had risen.

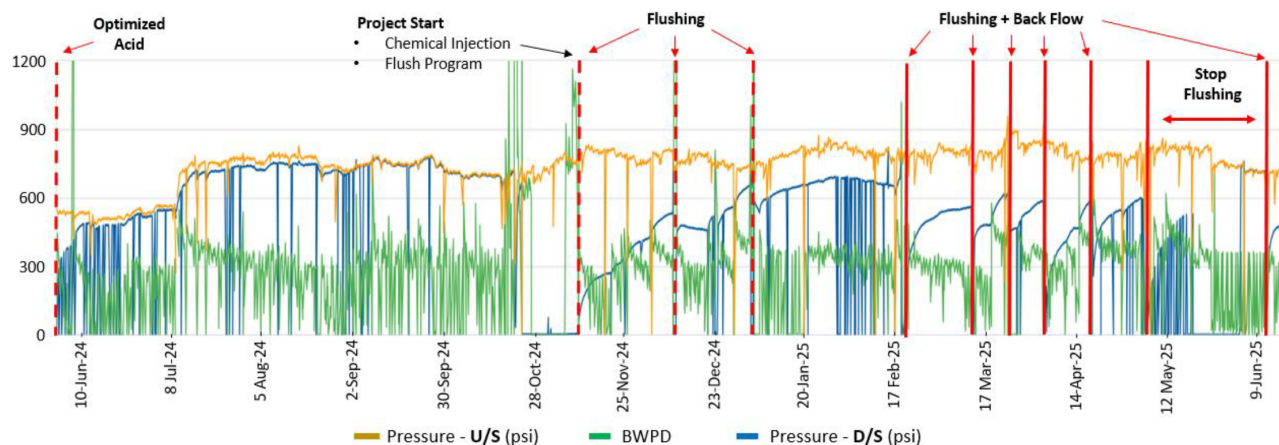


Figure 6—INJ-A1 Water Injection Performance post Optimized Acid CT Intervention

PWRI Wells INJ-A1, INJ-A2, INJ-A3 and INJ-A4

The flowline flushing and backflow sequence was performed on 3 other injector well candidates and the water injection performances are presented in Figures 7, 8 and 9 for wells INJ A1- A4 respectively.

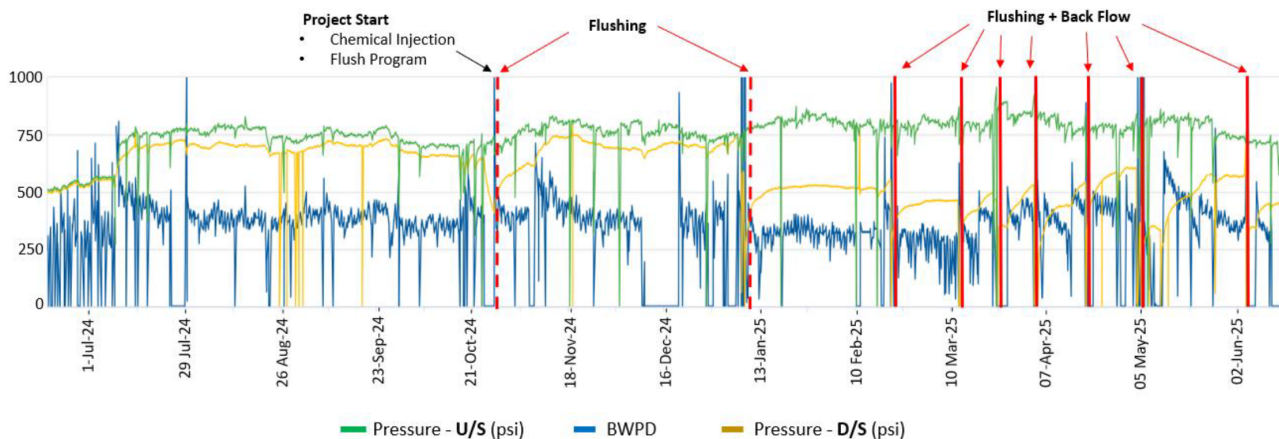


Figure 7—INJ-A2 Water Injection Performance

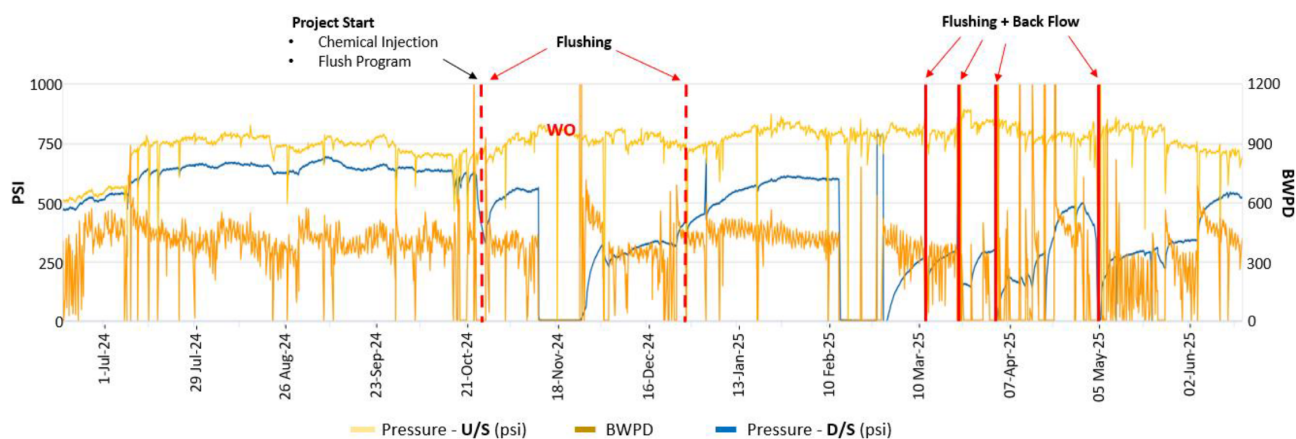


Figure 8—INJ-A3 Water Injection Performance

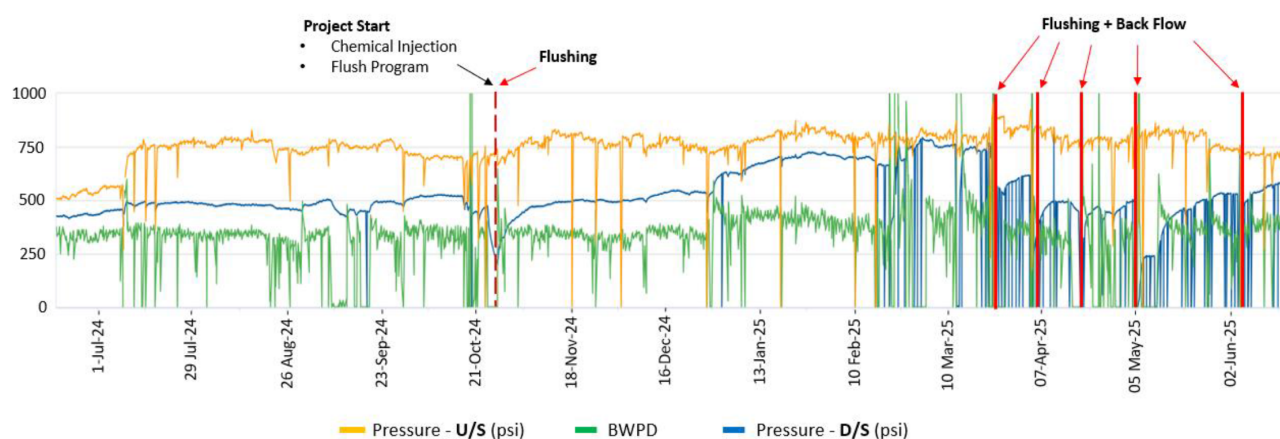


Figure 9—INJ-A4 Water Injection Performance

For PWRI well INJ-A2 the D/S pressures respond in a similar fashion to INJ-A1 following flowline flushes, however the water injection rate (blue line on this trend plot) shows a more positive cumulative response to successive flushes, see [Figure 7](#).

For PWRI well INJ-A3 the well experienced a CT-Optimized acid stimulation treatment in March 2024, and subsequently underwent a work over in November 2024. Scale inhibitor injection and first flushing operation occurred in October 2024. The D/S pressure (blue profile in [Figure 8](#)) and water injection rate (amber color profile) show impact of a more positive cumulative response to successive flushes, see [Figure 8](#).

PWRI well INJ-A4 initiated scale inhibitor injection and first flushing operation in October 2024. The D/S pressure (blue profile in [Figure 9](#)) and water injection rate (green color profile) show impact of a more positive cumulative response to successive flushes with overall lowering of D/S injection pressure, see [Figure 8](#).

Summary

Solid inorganic particle species including reservoir debris/fines and formed CaCO_3 (calcite), CaSO_4 (anhydrite) and FeS scale particles were confirmed by XRD to be present in the topsides PWRI system and are the main cause of blockage of pump inlet filters topsides and also blocking wells downhole in the PWRI wells reducing well injectivity performance.

The PWRI injectivity improvement team initiated remedial mineral acid application in 2023 to improve PWRI well injectivity via CT deployed mineral acid treatment spotted into the PWRI well near-wellbore

carbonate formations with varying success in recovering injectivity however the improvements were generally short-lived and injectivity decline observed. The team then initiated a two phase plan commencing 2024; firstly identify the causes of the mineral acid treatment failures and the nature of the blocking materials via intensive system sampling and analysis, and secondly implement a revised remedial strategy targeted at recovering PWRI injectivity to acceptable levels. A three pronged approach was decided which involved:

- i. Flushing - Decontamination of topside flowlines initially via flushing for 24 hours with sulphide scale dissolver in the water injection header, and PWRI well INJ-A2 flowline, and then periodic flushing of topside flowlines and backflowing the particle mix for discharge into a topsides disposal pool.
- ii. Topside scale suppression of via application of scale inhibitor upstream of the target PWRI cluster wells INJ-A1-A4.
- iii. Application of a revised optimized stimulation treatment recipe into the near wellbore of at-risk faltering PWRI injectors.

Sustained Success

In wells treated with the new stimulation recipe (INJ-A2 and INJ-A4) the injectivity performance was observed to improve and remain sustained, even though the monitored downstream injection pressure quickly increased post-stimulation treatment.

Pilot application of the scale inhibitor to topsides waters feeding the PWRI array exerted a beneficial effect on reducing PWRI blocking intensity by reducing the impact of anhydrite scale development topsides which had a knock-on effect downhole.

The scheduling of periodic flowline flushes for the Target PWRI wells and backflow of flushings into the storage pool topside greatly enhanced the uptime of the injectors by reducing the ‘clean and replace frequency of the topside pump inlet filters’ and also slowing of PWRI well injectivity performance decline.

By combination of the three components of the mitigation strategy, the uptime and injectivity performance of injectors to waterflood performance has been significant, and average BWPD injection rate has increased from 300 BWPD in July 2022 to approximately 1200 BWPD in 2025, resulting in an estimated increase in oil recovery of approximately 1170 BOPD, see [Figure 10](#) (however 50% is deducted as costs associated with CT unit daily costs, plus chemicals costs plus injection skid daily hire costs, therefore the net gain is 585BOPD).

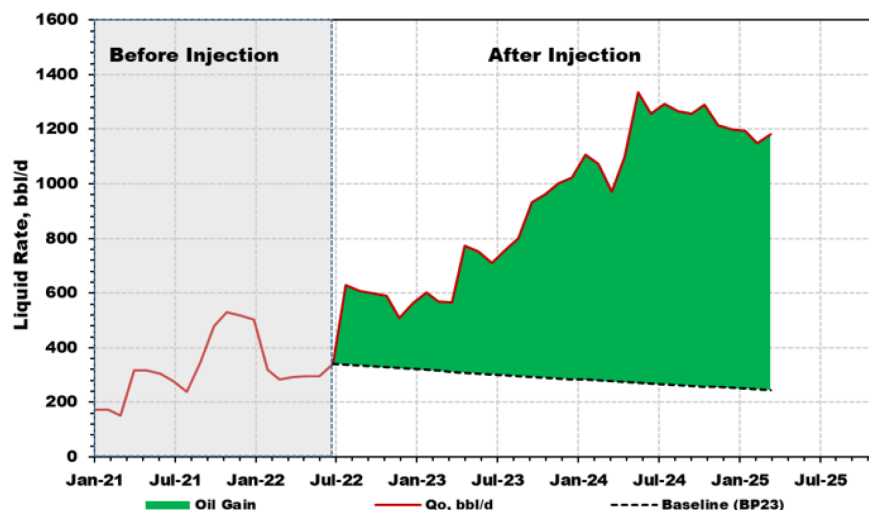


Figure 10—Study area waterflood Production

Conclusions

- Surface and subsurface samples were collected and analyzed, revealing the presence of sulfide and sulfate particles of concern, which were contributing factors to reduced injectivity in the water injection (WI) wells.
- The application of CaSO_4 scale inhibitor, combined with flowline flushing and backflow, significantly improved injectivity performance.
- Flowline dissolver flushing played a crucial role in enhancing PWRI performance by effectively removing iron sulfide deposits.
- The combined approach also helped reduce the need for costly well intervention activities (CT), thereby optimizing operational expenses.
- Overall, the injection project contributed to sustaining an oil gain of 585 BOPD from offset producers, demonstrating its effectiveness in improving field performance.

Way Forward

- Flushing to be undertaken on an ad-hoc basis.
- Continue scale inhibitor application in Pilot. Roll-out to other injection wells.
- Schedule periodic dosing of sulfide dissolver.
- Optimize scale inhibitor dosing and dosing locations for maximum benefit for entire waterflood system

Additional

Produced Water Clarity Before and After Scale Inhibitor Dosing topsides

In the Study Field water process, produced waters are commingled from a large number of well sources and channeled into a common system to feed the PWRI injector wells. The waters are of varying compatibilities and in bulk solution the compatibility issues increase and are aggravated by the presence of produced water inorganic suspended solids which can facilitate heterogeneous mineral scale formation.

Topside PWRI injector well filters tend to foul frequently with a thick sticky dark filter cake which can build quickly causing flow restriction and eventual blockage. The filters are routinely taken offline and cleaned for re-use.

Injection water feed samples were taken from various flowline locations in sequence from upstream flowline to side-stream sample after the pumps, see [Figures 11 to 15](#). The water sample taken furthest upstream ([Figure 11](#)) shows clear evidence of haze development suggesting scaling. At the pump inlet ([Figure 12](#)) the haziness was visibly worse, and solid precipitate could be seen on the bottle base. Samples recovered after the filters ([Figure 13](#)) are very hazy, and sample taken after the pump via side-stream sampler show hazing ([Figure 14](#)).



Figure 11—Upstream Flowline water samples showing hazing suggesting scaling



Figure 12—At pump inlet, showing increased haziness and solid deposits at the base of the bottles.



Figure 13—Water samples after the filters

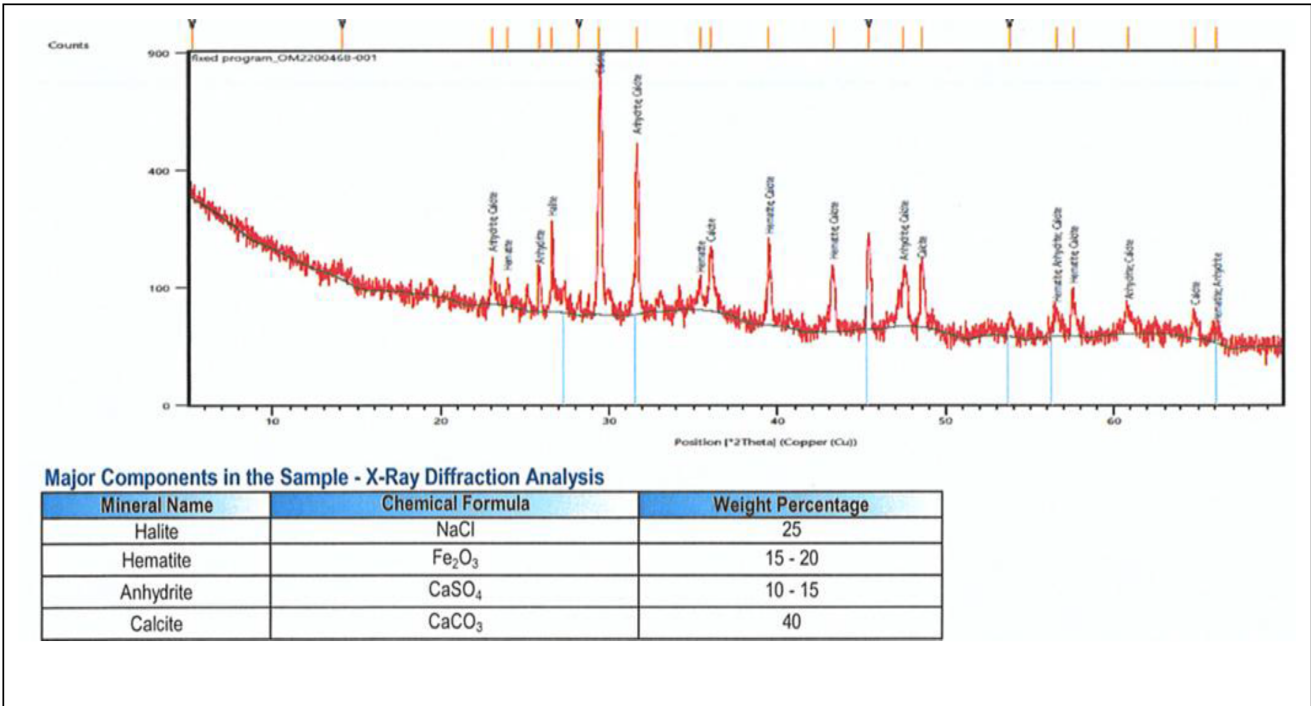


Figure 14—sample obtained via side-stream after pumps

Produced Water Sample Solids Analysis

Solids collected from the flowlines, filters and pumps surface also contain high amounts of sulfate see Table 1 for the XRD and elemental weight percent for the solids recovered from Location 1 produced water sample, and samples recovered from all Locations 1 – 3 XRD results presented in Table 2.

Table 1—XRD of solids collected at injection flowline



The sulfate scale inhibitor was introduced into the line to inhibit sulfate scaling in the topside flowlines and the result was a less hazy water since scale formation was inhibited making the water much clearer and cleaner. Figure 15–17 shows cleaner water samples from the flowlines, filter and pumps side streams.



Figure 15—Flowline samples - no haziness after treatment.



Figure 16—samples taken before the pumps, clean and clear



Figure 17—Sample from side-stream after pump are haze free.

Reference

1. Al-Mufargi, H., Al-Harhi, H., Al-Naabi, A., Al-Shuelli, M., Al-Wahaibi, K., 2025. The Journey and Evaluation of Chemical Water Shut-Off Treatments in Carbonate Reservoirs: A Case Study from Northern Oman. *SPE Oman Petroleum & Energy Show*, 12-14 May, 2025, Muscat, Oman, SPE-224978-MS
2. Zhang, L., Zhou, L., Al-Mugheiry, M.A., Xu, K., 2007. Horizontal Waterflooding in Shuaiba Carbonate Reservoir of Study Field in Oman: From Pilots' Performance To Development Era. SPE-108392-MS.
3. Dandan, H., Yuanbing, Wu., Zhang, H., Ma, R., Qian Yao, L., Songhao, H., Yihang, C., 2024. Field Application of Evaluation Model for Development Effect on Horizontal Waterflooding in Thin Low-permeability Carbonate Reservoirs. *J.Phys.Conf.Series* **2834**(1):012179.
4. Al-Harrasi, S., Al-Abri, A.T., Al-Badi, S., Hadidi, K., Al-Faraji, O., Al-Dafai, H., Al-Busaidi, K., Al-Wahaibi, K., Al-Abri, A.N., Khemnari, A., 2025. Deepening of Sucker-Rod Pumps in Horizontal Wells to Boost Oil Production. *SPE Oman Petroleum & Energy Show*, Muscat, Oman, May 2025. SPE-224843-MS.
5. Baveye, P., Vandevivere, P., Hoyle, B. L., DeLeo, P. C., and de Lozada, D. S. 1998. Environmental Impact and Mechanisms of the Biological Clogging of Saturated Soils and Aquifer Materials. *Crit. Rev. Environ. Sci. Technol.* **28** (2): 123–191.
6. Bennion, D. B., 2002. An Overview of Formation Damage Mechanisms Causing a Reduction in Productivity and Injectivity of Oil and Gas Producing Formations. *J. Can. Pet. Technol.* **41** (11).
7. Feia, S., Dupla, J. C., Ghabezloo, S., Sulem, J., Canou, J., Onaisi, A., et al 2015. Experimental Investigation of Particle Suspension Injection and Permeability Impairment in Porous Media. *Geomech. Energy Environ.* **3**: 24–39.
8. Hearn, C. L., and Whitson, C. 1995. Evaluating Miscible and Immiscible Gas Injection in the Sarah Field, Oman. Presented at the 13th SPE Symposium on Reservoir Simulation, San Antonio, Texas, 6–8 February. SPE-29115-MS.
9. Hua, G. F., Zhu, W., Zhao, L. F., and Huang, J. Y. 2010. Clogging Pattern in Vertical Flow Constructed Wetlands: Insight from a Laboratory Study. *J. Hazard. Mater.* **180** (1–3): 668–674.
10. Jeong, H. Y., Jun, S. C., Cheon, J. Y., and Park, M. A. 2018. Review on Clogging Mechanisms and Managements in Aquifer Storage and Recovery (ASR) Applications. *Geosci. J.* **22**: 667–679.
11. Kazmi, S., Mushir, K., Khan, A., Ali, M., Saqib, C., Shaheed, B., AbdelGhaffar, M., and Al Suqri, H. 2024. Reservoir Characterization of Carbonate Reservoir by Heterogeneity Analysis Using High-Definition Borehole Image & Petrophysical Data of Upper Shuaiba Formation Study Field, Oman. Presented at the Abu Dhabi International Petroleum Exhibition and Conference, Abu Dhabi, UAE, November. SPE-222079-MS.
12. Moghadasi, J., Müller-Steinhagen, H., Jamialahmadi, M., and Sharif, A. 2004. Theoretical and Experimental Study of Particle Movement and Deposition in Porous Media During Water Injection. *J. Pet. Sci. Eng.* **43** (3–4): 163–181.
13. Rinck-Pfeiffer, S., Ragusa, S., Sztajn bok, P., and Vandeveld, T. 2000. Interrelationships Between Biological, Chemical, and Physical Processes as an Analog to Clogging in Aquifer Storage and Recovery (ASR) Wells. *Water Res.* **34** (7): 2110–2118.
14. Song, W., Liu, X., Zheng, T., and Yang, J. 2020. A Review of Recharge and Clogging in Sandstone Aquifer. *Geothermics* **87**: 101857.
15. Wang, W., Wei, W., Ferrier, N., and Arismendi, N. 2018. Scale Formation and Inhibition Study for Water Injection Wells. Presented at the SPE International Oilfield Scale Conference and Exhibition, Aberdeen, Scotland, UK, June. SPE-190732-MS.

16. Wang, Z., Du, X., Yang, Y., and Ye, X. 2012. Surface Clogging Process Modeling of Suspended Solids During Urban Stormwater Aquifer Recharge. *J. Environ. Sci. (China)* **24** (8): 1418–1424.
17. Yuan, B., and Wood, D. A., 2018. A Comprehensive Review of Formation Damage During Enhanced Oil Recovery. *J. Pet. Sci. Eng.* **167**: 287–299.